

Betting-against-Beta and Volatility Factors

Figure 10.13 compares the *BAB* and *VOL* factors from October 1963 to December 2011.²⁸ The cumulative returns of the *VOL* factor are higher than *BAB*, and the volatility factor has a slightly higher Sharpe ratio (0.6 vs. 0.5), but the two factors are largely comparable. The main surprising result is that the beta and volatility effects are very lowly correlated; the correlation between *BAB* and *VOL* is -9% . The volatility and beta anomalies, therefore, are distinct.

Running a Fama–French plus momentum factor regression, we obtain

	BAB Factor		VOL Factor	
	Coeff	T-stat	Coeff	T-stat
Alpha	0.33%	1.89	0.42%	4.37
MKT Loading	-0.17	-4.13	0.87	38.8
SMB Loading	0.29	5.20	-0.63	-20.3
HML Loading	0.48	7.85	0.20	5.73
UMD Loading	0.09	2.35	0.13	6.00

The alpha for the *BAB* factor is 0.33% per month (4% per year) and the t -statistic of 1.89 corresponds to a p -value of 0.06. So this is borderline statistically significant at the standard 95% level. The *VOL* factor’s alpha is slightly

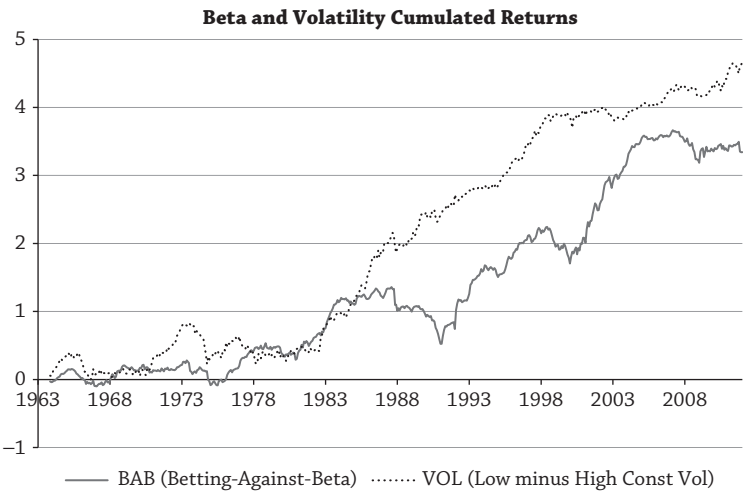


Figure 10.13

²⁸I construct a *BAB* factor similar to Frazzini and Pedersen (2012) except I do not follow their step in shrinking the betas. Specifically, betas are computed in one-year rolling regressions using daily frequency returns. There are two beta portfolios created at the end of each month, and the *BAB* factor is constructed using equation (10.14) using the pre-ranking portfolio betas.